BRADLEY NAST

+27 81 017 6273 or +27 83 642 8790 | bradleypnast@gmail.com
linkedin.com/in/bradleynast | github.com/0xBPNast/Portfolio

SUMMARY

Data Scientist / ML Engineer with an engineering background and 5+ years delivering analytics-driven technology projects in the mining industry. Experienced in translating operational problems into analytical tasks, building and evaluating ML models in Python, and delivering production-minded solutions (automation, APIs, containers). Strong foundation in statistics, time series, and stakeholder communication.

SKILLS

Programming & Analytics	Python (pandas, numpy, scikit-learn, matplotlib), SQL, Git, Jupyter, Power BI, Excel
Machine Learning	Classification/regression, supervised & unsupervised learning, feature engineering, model evaluation, cross-validation, error analysis
ML Engineering	API-based deployment (FastAPI), Docker, pipeline design, experiment structure, version control
Math & Stats	Probability, linear algebra, optimization, time series analysis
Certifications	AWS Machine Learning Engineer Specialty (In Progress), Deep Learning Specialization (Coursera – In Progress)

EDUCATION

University of Pretoria, Pretoria, Gauteng — Dec 2024
Bachelor of Engineering Honours, Mechanical Engineering with Distinction (NQF Level 8)
- Research: Trained a 1D-CNN on time-series vibration data for fault detection and classification.
- Golden Key Society.

Waterstone College, Johannesburg, Gauteng — Dec 2014
Matric Certificate (NQF Level 4)

PROJECTS

Predicting Equipment Failures with Time Series Data | 2025
- Framed equipment failure risk as a supervised learning problem; developed a 1D-CNN model to estimate failure probability from telemetry/synthetic time-series signals.
- Engineered time- and frequency-domain features (FFT, RMS, kurtosis) across 1M+ datapoints; used validation splits appropriate for time-series to reduce leakage risk.
- Evaluated model performance using classification metrics (e.g., precision/recall/F1) and iterated on features/architecture based on error analysis.

ML-Powered Maintenance Classifier | 2024
- Labeled 200+ maintenance logs; built an NLP preprocessing + classification pipeline to categorize issue types for improved reporting consistency.
- Presented outputs in Power BI to demonstrate operational usability and enable stakeholders to identify recurring failure themes.

EXPERIENCE

Product Manager - Automation
Epiroc, Johannesburg, Gauteng — June 2025 – Present
- Built a consolidated database of disparate data for use in downstream analytics.

Project Engineer
Epiroc, Johannesburg, Gauteng — Oct 2022 – Present
- Partnered with technical and operational stakeholders to scope condition-monitoring needs and translate requirements into data capture, analytical workflows, and reporting outputs.

- Designed and deployed condition-monitoring solutions using real-time data capture and visualization to improve asset health visibility and support proactive maintenance.
- Conducted statistical analysis on telemetry data in Python/Excel to identify trends/anomalies and support maintenance decision-making.
- Integrated data workflows from disparate sources (e.g., PLC logs, CSV extracts) to automate reporting pipelines and reduce manual effort.
- Led an internal ML pilot to cluster maintenance event types using unsupervised learning, improving consistency in categorization and follow-up actions.

Engineer-in-Training
Epiroc, Johannesburg, Gauteng Jan 2021 – Sep 2022

- Built Excel macros to automate recurring internal processes and improve data quality/throughput.
- Recognized for Service & Operational Excellence for delivery performance and customer impact.
- Promoted within 6 months due to strong performance and organizational impact (6 months ahead of schedule).